

Лабораторная работа №13

Фильтр пакетов

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Информация

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Цели и задачи

- Получить навыки настройки пакетного фильтра в Linux.
1. Используя `firewall-cmd`:
 - определить текущую зону по умолчанию;
 - определить доступные для настройки зоны;
 - определить службы, включённые в текущую зону;
 - добавить сервер VNC в конфигурацию брандмауэра.
 2. Используя `firewall-config`:
 - добавьте службы `http` и `ssh` в зону `public`;
 - добавьте порт 2022 протокола UDP в зону `public`;
 - добавьте службу `ftp`.
 3. Выполните задание для самостоятельной работы (раздел 13.5).

Материалы и методы

- Linux (дистрибутив Rocky 9.6)
- Linux Fedora Sway (Markdown)
- Oracle VirtualBox

Выполнение лабораторной работы

Управление брандмауэром с помощью firewall-cmd

```
firewall-cmd --get-default-zone
```

```
firewall-cmd --get-zone
```

```
firewall-cmd --get-services
```

```
firewall-cmd --list-services
```

```
[aeakunaeva@aeakunaeva ~]$ su -
Password:
[root@aeakunaeva ~]# firewall-cmd --get-default-zone
public
[root@aeakunaeva ~]# firewall-cmd --get-zones
block dmz drop external home internal nm-shared public trusted work
[root@aeakunaeva ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit auswei
sapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc b
itcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent coc
kpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client dist
cc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server f
inger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp g
alera ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap i
maps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klog
in kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-contro
ller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llm
nr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt m
qtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nu
t opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pm
webapi pop3 pop3s postgresql proxyo prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netrv p
p pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rts
r salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmp
tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing sync
thing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmiss
ion-client upnp-client vdsim vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discove
ry-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-ser
ver zabbix-agent zabbix-server zerotier
[root@aeakunaeva ~]# firewall-cmd --list-services
cockpit dhcpv6-client ssh
```

Вывод информации в конфигурации и добавление служб в firewalld

```
firewall-cmd --list-all
```

```
firewall-cmd --list-all --zone=public
```

```
firewall-cmd --add-service=vnc-server
```

```
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@aeakunaeva ~]# firewall-cmd --list-all --zone=public
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
```

Добавление и проверка службы в конфигурации

```
firewall-cmd --add-service=vnc-server --permanent
```

```
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@aeakunaeva ~]# systemctl restart firewalld
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
```

Проверка наличия постоянной службы vnc-server

```
firewall-cmd --reload
```

```
firewall-cmd --list-all
```

```
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@aeakunaeva ~]# firewall-cmd --reload
success
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
```

Добавление порта в конфигурацию

```
firewall-cmd --add-port=2022/tcp --permanent
```

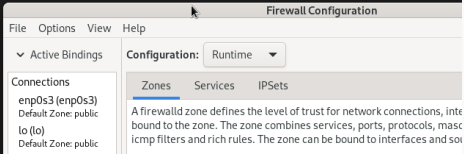
```
[root@aeakunaeva ~]# firewall-cmd --add-port=2022/tcp --permanent
success
[root@aeakunaeva ~]# firewall-cmd --reload
success
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports: 2022/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

firewall-config

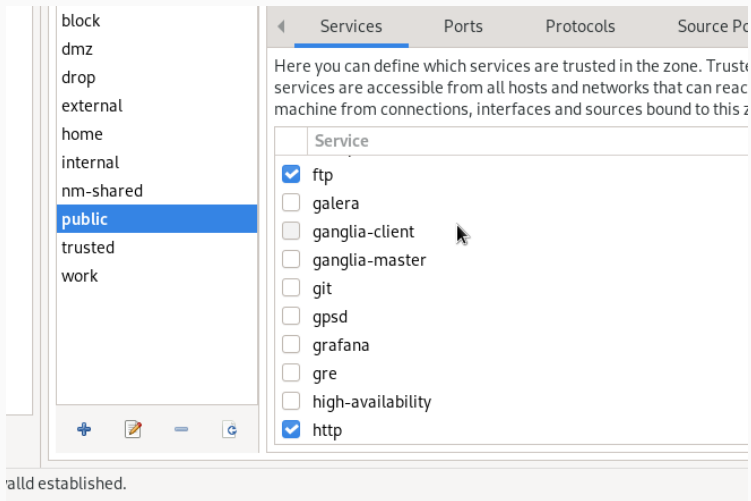
```
[root@aeakunaeva ~]# GNU firewall-config: firewall-config
bash: GNU: command not found...
[root@aeakunaeva ~]# firewall-config
bash: firewall-config: command not found...
Install package 'firewall-config' to provide command 'firewall-config'? [N/y] y
```

```
* Waiting in queue...
* Loading list of packages....
The following packages have to be installed:
dbus-x11-1:1.12.20-8.el9.x86_64      X11-requiring add-ons for D-BUS
firewall-config-1.3.4-15.el9.noarch Firewall configuration application
Proceed with changes? [N/y] y
```

```
* Waiting in queue...
* Waiting for authentication..
* Waiting in queue...
* Downloading packages...
* Requesting data...
* Testing changes...
* Installing packages...
```



Добавление служб в firewall-config



block
dmz
drop
external
home
internal
nm-shared
public
trusted
work

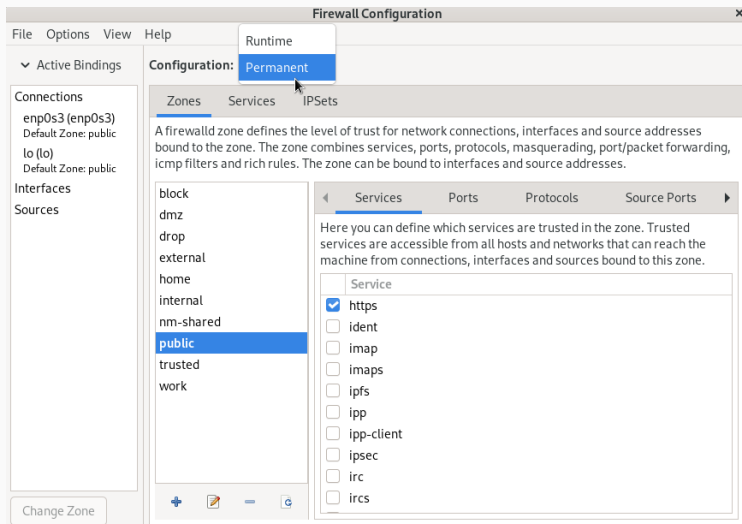
Services Ports Protocols Source Ports

Here you can define which services are trusted in the zone. Trusted services are accessible from all hosts and networks that can reach the machine from connections, interfaces and sources bound to this zone.

Service
☒ ftp
☐ galera
☐ ganglia-client
☐ ganglia-master
☐ git
☐ gpsd
☐ grafana
☐ gre
☐ high-availability
☒ http

all established.

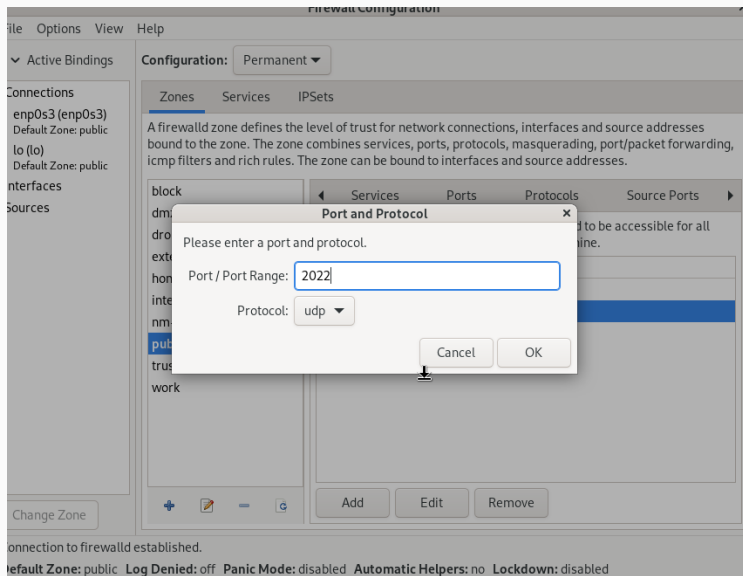
Выбор настройки конфигурации в firewall-config



Connection to firewalld established.

Default Zone: public Log Denied: off Panic Mode: disabled Automatic Helpers: no Lockdown: disabled

Добавление порта в firewall-config



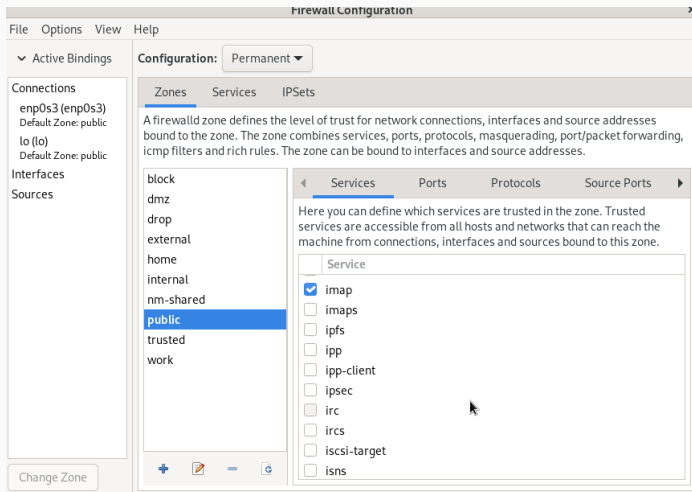
Проверка изменений в интерфейсе firewall-config

```
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports: 2022/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@aeakunaeva ~]# firewall-cmd --reload
success
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https ssh vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

```
firewall-cmd --add-service=telnet --permanent
```

```
[root@aeakunaeva ~]# firewall-cmd --add-service=telnet --permanent  
success
```

firewall-config



```
firewall-cmd --reload
```

```
firewall-cmd --list-all
```

```
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https ssh vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@aeakunaeva ~]# firewall-cmd --reload
success
[root@aeakunaeva ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https imap pop3 smtp ssh telnet vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
```

Выводы

Я получила навыки настройки пакетного фильтра в Linux.